PROJECT ATLAS

Autonomous Revenue Operating System — AgenThink Mesh

Architecture & Implementation Document v1.0

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> **THE BET:** AgenThink Mesh can reach $20M ARR with 8 humans if and only if the revenue engine is autonomous, auditable, and continuously improving — and if every engineering decision is subordinated to customer acquisition, not agent sophistication.

EXECUTIVE SUMMARY

Project ATLAS is the architecture for the Autonomous Revenue Operating System (AROS) of AgenThink Mesh. It defines how the company will discover, qualify, engage, and close 30+ enterprise customers across 9 geographies and 10 sectors using 200+ specialised agents coordinated by 8 human operators.

The architecture is organised around five production systems, six phases, and one governing principle: **every module must answer “Does this help acquire, retain, or expand customers?” If not, it is deferred.**

The end state is a commercial machine that operates at a scale no human team of 8 could reach manually: 100,000 companies indexed, 10,000 executive profiles maintained, 1,000 active opportunities tracked, and $20M+ ARR at 85%+ gross margins.

PART 1 — SYSTEM ARCHITECTURE

1.1 Overview

The Autonomous Revenue Operating System consists of five production systems operating in a continuous loop:

```
DISCOVERY → INTELLIGENCE → DETECTION → OUTREACH → COMMAND
    ↑                                                  |
    └────────────── LEARNING FEEDBACK LOOP ────────────┘
```

Each system is independently deployable, independently scalable, and independently auditable. They share a common data layer, a common agent orchestration framework, and a common token accounting system.

1.2 System Inventory

System	Name	Purpose	Agent Swarm	Output
S1	Global Target Discovery Engine	Continuously discover enterprise targets	Discovery Swarm	targets.csv
S2	Account Intelligence Factory	Build 5–10 page account dossiers	Intelligence Swarm	account*dossier*[company].md
S3	Decision Detection Engine	Identify strategic decisions before competitors	Decision Swarm	decision_pipeline.csv
S4	Outreach Factory	Generate personalised outreach campaigns	Outreach Swarm + Proposal Swarm	outreach_queue.csv
S5	Revenue Command Center	Executive dashboard and pipeline management	Finance Swarm + Governance Swarm	Live dashboard

1.3 Data Flow Architecture

```
External Sources (web, news, filings, LinkedIn, databases)
        │
        ▼
[S1: Discovery Engine] ──────────────────────────────────────┐
        │ targets.csv                                        │
        ▼                                                    │
[S2: Intelligence Factory] ─────────────────────────────────┤
        │ account_dossier_[company].md                       │
        ▼                                                    │
[S3: Decision Detection] ───────────────────────────────────┤
        │ decision_pipeline.csv                              │
        ▼                                                    │
[S4: Outreach Factory] ─────────────────────────────────────┤
        │ outreach_queue.csv (pending human approval)        │
        ▼                                                    │
[Human Approval Gate] ◄──────────────────────────────────────┘
        │ approved_outreach.csv
        ▼
[CRM / Engagement Tracking]
        │ outcomes.csv
        ▼
[S5: Revenue Command Center]
        │ performance_metrics.json
        ▼
[Learning Feedback Loop] ──────────► back to S1 targeting criteria
```

1.4 Integration Layer

The five systems communicate through a shared PostgreSQL database (primary data store), an S3-compatible object store (dossiers, proposals, reports), a Redis message queue (agent task coordination), and a REST API gateway (human operator interfaces).

All agent outputs are written to the database before being consumed by downstream systems. No agent communicates directly with another agent — all coordination is mediated through the data layer. This is the primary architectural decision that enables auditability.

1.5 Human Touchpoints

Human operators interact with the system at exactly four points:

1. **Outreach approval queue** — review and approve/reject outreach before sending
2. **Opportunity qualification** — confirm or reject AI-scored opportunities above threshold
3. **Proposal review** — review AI-generated proposals before delivery

4. **System configuration** — adjust targeting criteria, scoring thresholds, agent parameters

Everything else is autonomous.

PART 2 — DATABASE SCHEMA

2.1 Core Tables

The schema is organised into six domains: Companies, Contacts, Decisions, Outreach, Pipeline, and Agent Operations.

Domain 1: Companies

```
CREATE TABLE companies (
  id                      UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  name                    TEXT NOT NULL,
  domain                  TEXT UNIQUE,
  headquarters_country    TEXT,
  headquarters_city       TEXT,
  sector                  TEXT,  -- enum: Banking, AssetMgmt, Insurance, Telecom, I
  region                  TEXT,  -- enum: Americas, EMEA, APAC
  revenue_estimate_usd    BIGINT,
  employee_count          INTEGER,
  digital_transform_score INTEGER CHECK (digital_transform_score BETWEEN 0 AND 10
  ai_adoption_score       INTEGER CHECK (ai_adoption_score BETWEEN 0 AND 100),
  opportunity_score       INTEGER CHECK (opportunity_score BETWEEN 0 AND 100),
  estimated_acv_usd       INTEGER,
  estimated_buying_capacity TEXT,  -- Low / Medium / High / Very High
  dossier_status          TEXT DEFAULT 'pending',  -- pending / in_progress / compl
  last_enriched_at        TIMESTAMPTZ,
  created_at              TIMESTAMPTZ DEFAULT now(),
  updated_at              TIMESTAMPTZ DEFAULT now()
);

CREATE TABLE company_signals (
  id             UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  company_id     UUID REFERENCES companies(id),
  signal_type    TEXT,  -- ai_adoption / digital_transform / m_and_a / leadership_
  signal_text    TEXT,
  source_url     TEXT,
  signal_date    DATE,
  confidence     INTEGER CHECK (confidence BETWEEN 0 AND 100),
  created_at     TIMESTAMPTZ DEFAULT now()
);

CREATE TABLE company_dossiers (
  id               UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  company_id       UUID REFERENCES companies(id),
  version          INTEGER DEFAULT 1,
  executive_brief TEXT,
  strategic_challenges TEXT,
  competitive_threats TEXT,
  ai_readiness_assessment TEXT,
  decision_twin    TEXT,
  historical_analogues TEXT,
  mesh_use_cases   TEXT,
  opportunity_score INTEGER,
  outreach_strategy TEXT,
```

```sql
  full_markdown   TEXT,  -- S3 key to full .md file
  generated_at    TIMESTAMPTZ DEFAULT now(),
  approved_by     UUID REFERENCES users(id),
  status          TEXT DEFAULT 'draft'  -- draft / approved / archived
);
```

Domain 2: Contacts

```sql
CREATE TABLE contacts (
  id                UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  company_id        UUID REFERENCES companies(id),
  full_name         TEXT NOT NULL,
  title             TEXT,
  seniority         TEXT,  -- C-Suite / VP / Director / Manager
  function          TEXT,  -- CEO / CMO / CTO / CDO / CIO / CFO / Head-Clinical / H
  linkedin_url      TEXT,
  email             TEXT,
  email_verified    BOOLEAN DEFAULT false,
  email_source      TEXT,
  phone             TEXT,
  location          TEXT,
  last_verified_at TIMESTAMPTZ,
  created_at        TIMESTAMPTZ DEFAULT now()
);
```

Domain 3: Decisions

```sql
CREATE TABLE decisions (
  id                      UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  company_id              UUID REFERENCES companies(id),
  decision_type           TEXT,  -- MA / DigitalTransform / AIAdoption / ETFLaunch
  decision_brief          TEXT,
  stakeholder_map         TEXT,
  decision_twin           TEXT,
  estimated_value_usd     BIGINT,
  engagement_probability INTEGER CHECK (engagement_probability BETWEEN 0 AND 100)
  source_url              TEXT,
  detected_at             TIMESTAMPTZ DEFAULT now(),
  status                  TEXT DEFAULT 'active'  -- active / engaged / closed / exp
);
```

Domain 4: Outreach

```
CREATE TABLE outreach_campaigns (
  id               UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  company_id       UUID REFERENCES companies(id),
  contact_id       UUID REFERENCES contacts(id),
  decision_id      UUID REFERENCES decisions(id),
  campaign_type    TEXT,  -- intro_email / linkedin / followup_1 / followup_2 / ex
  subject_line     TEXT,
  body_text        TEXT,
  personalisation_notes TEXT,
  status           TEXT DEFAULT 'pending_approval',  -- pending_approval / approve
  approved_by      UUID REFERENCES users(id),
  approved_at      TIMESTAMPTZ,
  sent_at          TIMESTAMPTZ,
  replied_at       TIMESTAMPTZ,
  outcome          TEXT,  -- meeting_booked / not_interested / no_response / bounc
  created_at       TIMESTAMPTZ DEFAULT now()
);
```

Domain 5: Pipeline

```
CREATE TABLE opportunities (
  id                  UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  company_id          UUID REFERENCES companies(id),
  contact_id          UUID REFERENCES contacts(id),
  stage               TEXT,  -- discovered / dossier_complete / decision_detectec
  acv_estimate_usd    INTEGER,
  probability         INTEGER CHECK (probability BETWEEN 0 AND 100),
  expected_close_date DATE,
  account_director    UUID REFERENCES users(id),
  notes               TEXT,
  created_at          TIMESTAMPTZ DEFAULT now(),
  updated_at          TIMESTAMPTZ DEFAULT now()
);

CREATE TABLE proposals (
  id              UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  opportunity_id  UUID REFERENCES opportunities(id),
  version         INTEGER DEFAULT 1,
  title           TEXT,
  executive_summary TEXT,
  use_case        TEXT,
  pricing_model   TEXT,
  pilot_scope     TEXT,
  full_markdown   TEXT,  -- S3 key
  status          TEXT DEFAULT 'draft',  -- draft / approved / delivered / accept
  delivered_at    TIMESTAMPTZ,
  created_at      TIMESTAMPTZ DEFAULT now()
);
```

Domain 6: Agent Operations

```
CREATE TABLE agent_runs (
  id               UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  agent_name       TEXT NOT NULL,
  swarm            TEXT,  -- Discovery / Intelligence / Decision / Outreach / Prop
  task_type        TEXT,
  input_summary    TEXT,
  output_summary   TEXT,
  tokens_consumed  INTEGER,
  cost_usd         NUMERIC(10,6),
  duration_ms      INTEGER,
  status           TEXT,  -- success / failure / timeout / throttled
  error_message    TEXT,
  revenue_influenced_usd BIGINT DEFAULT 0,
  roi_score        NUMERIC(10,4),  -- revenue_influenced / cost_generated
  created_at       TIMESTAMPTZ DEFAULT now()
);

CREATE TABLE agent_throttle_log (
  id           UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  agent_name   TEXT,
  reason       TEXT,  -- low_roi / cost_overrun / rate_limit / manual
  throttled_at TIMESTAMPTZ DEFAULT now(),
  restored_at TIMESTAMPTZ,
  throttled_by TEXT  -- system / human
);
```

2.2 Key Indexes

```
CREATE INDEX idx_companies_sector ON companies(sector);
CREATE INDEX idx_companies_region ON companies(region);
CREATE INDEX idx_companies_opportunity_score ON companies(opportunity_score DESC)
CREATE INDEX idx_decisions_type ON decisions(decision_type);
CREATE INDEX idx_decisions_detected_at ON decisions(detected_at DESC);
CREATE INDEX idx_outreach_status ON outreach_campaigns(status);
CREATE INDEX idx_opportunities_stage ON opportunities(stage);
CREATE INDEX idx_agent_runs_swarm ON agent_runs(swarm);
CREATE INDEX idx_agent_runs_roi ON agent_runs(roi_score DESC);
```

PART 3 — AZURE ARCHITECTURE

3.1 Service Map

The AROS runs on Azure with the following service configuration:

Service	Azure Product	Purpose	Tier
Primary Database	Azure Database for PostgreSQL Flexible Server	All structured data	General Purpose, 4 vCores
Object Storage	Azure Blob Storage	Dossiers, proposals, reports	LRS, Hot tier
Message Queue	Azure Service Bus	Agent task coordination	Standard
API Gateway	Azure API Management	Human operator interfaces	Developer
Agent Compute	Azure Container Apps	Agent execution environment	Consumption plan
LLM Gateway	Azure OpenAI Service	All LLM calls	GPT-4o deployment
Search	Azure AI Search	Company and contact search	Basic
Monitoring	Azure Monitor + Application Insights	Performance, cost, errors	Standard
Key Vault	Azure Key Vault	Secrets management	Standard
CDN	Azure Front Door	Dashboard delivery	Standard

3.2 Agent Compute Architecture

All agents run as Azure Container Apps in the Consumption plan. This means:

- **Zero cost when idle** — agents only consume compute when processing tasks
- **Automatic scaling** — up to 300 replicas per agent type during peak discovery runs
- **Isolated execution** — each agent run is a separate container instance
- **Built-in retry** — failed tasks are automatically retried via Service Bus dead-letter queues

3.3 Cost Architecture

The primary cost driver is LLM tokens, not compute. The architecture is designed to minimise token consumption through three mechanisms:

1. **Tiered enrichment** — companies are enriched in stages (basic → signal → dossier) with human approval gates between stages. Only approved companies proceed to the next tier.
2. **Caching** — all LLM outputs are cached for 30 days. Repeated queries about the same company return cached results.
3. **Model routing** — simple classification tasks (sector assignment, signal scoring) use GPT-4o-mini. Complex generation tasks (dossiers, Decision Twins, proposals) use GPT-4o. Cost difference: approximately 15x.

3.4 Estimated Monthly Azure Spend

Component	Monthly Cost (100 companies/day)	Monthly Cost (1,000 companies/day)
PostgreSQL	$180	$360
Blob Storage	$20	$80
Service Bus	$10	$40
Container Apps	$150	$600
Azure OpenAI (GPT-4o)	$1,200	$8,000
Azure OpenAI (GPT-4o-mini)	$80	$400
AI Search	$75	$250
Monitoring	$50	$150
Total	**$1,765/month**	**$9,880/month**

At $9,880/monthinfrastructurecostand$20M ARR target, infrastructure is approximately 0.6% of revenue — well within the 85% gross margin target.

PART 4 — AGENT ARCHITECTURE

4.1 Agent Design Principles

Every agent in the AROS follows four design rules:

1. **Single responsibility** — each agent does exactly one thing
2. **Deterministic inputs** — agents receive structured inputs from the database, not free-form instructions
3. **Auditable outputs** — all outputs are written to the database with full provenance
4. **Token-accounted** — every agent call records tokens consumed, cost, and revenue influenced

4.2 Agent Swarm Inventory

Discovery Swarm (8 agents)

Agent	Task	Input	Output	Model
SectorScanner	Identify companies in target sector/region	Sector + Region	company_name, domain, country	GPT-4o-mini
CompanyProfiler	Extract basic company data	company_name	revenue, employees, HQ	GPT-4o-mini
AISignalDetector	Detect AI adoption signals	company_name, domain	signal_type, signal_text, confidence	GPT-4o-mini
DigitalTransformScorer	Score digital transformation activity	company signals	digital_transform_score	GPT-4o-mini
ExecutiveExtractor	Extract C-suite names and titles	company_name	contact list	GPT-4o-mini
OpportunityScorer	Score overall opportunity	company profile + signals	opportunity_score, estimated_acv	GPT-4o
DuplicateChecker	Prevent duplicate company entries	company_name, domain	is_duplicate, existing_id	GPT-4o-mini
QualityGatekeeper	Validate company data completeness	company record	pass/fail + missing fields	GPT-4o-mini

Intelligence Swarm (6 agents)

Agent	Task	Input	Output	Model
ExecutiveBriefWriter	Write executive brief section	company profile + signals	executive_brief (500 words)	GPT-4o
StrategicChallengesAnalyst	Identify strategic challenges	executive brief + sector context	strategic_challenges (400 words)	GPT-4o
CompetitiveThreatMapper	Map competitive threats	company profile + sector	competitive_threats (400 words)	GPT-4o
AIReadinessAssessor	Assess AI readiness	signals + digital score	ai_readiness_assessment (300 words)	GPT-4o
DecisionTwinBuilder	Build Decision Twin profile	executive brief + challenges	decision_twin (600 words)	GPT-4o
DossierAssembler	Assemble full dossier markdown	all sections	account*dossier*[company].md	GPT-4o-mini

Decision Swarm (5 agents)

Agent	Task	Input	Output	Model
NewsMonitor	Monitor for strategic news	company_name, domain	raw news items	GPT-4o-mini
DecisionClassifier	Classify decision type	news item	decision_type, confidence	GPT-4o-mini
StakeholderMapper	Map decision stakeholders	decision + company contacts	stakeholder_map	GPT-4o
DecisionTwinMatcher	Match decision to Decision Twin	decision + dossier	decision_twin, engagement_probability	GPT-4o
DecisionBriefWriter	Write decision brief	all decision data	decision_brief (300 words)	GPT-4o

Outreach Swarm (5 agents)

Agent	Task	Input	Output	Model
IntroEmailWriter	Write personalised intro email	dossier + decision + contact	intro_email (subject + body)	GPT-4o
LinkedInMessageWriter	Write LinkedIn outreach	dossier + contact	linkedin_message (300 chars)	GPT-4o
FollowUpSequenceWriter	Write follow-up 1 and 2	intro email + outcome	followup_1, followup_2	GPT-4o
ExecBriefingRequestWriter	Write executive briefing request	dossier + decision	exec_briefing_request	GPT-4o
SDRTeaserWriter	Write SDR teaser proposal	dossier + use cases	sdr_teaser (1 page)	GPT-4o

Proposal Swarm (4 agents)

Agent	Task	Input	Output	Model
UseCaseSelector	Select best Mesh use cases	dossier + decision	top 3 use cases	GPT-4o
ProposalWriter	Write full commercial proposal	use cases + pricing	proposal.md	GPT-4o
PricingModeller	Generate pricing options	company size + use case	pricing_model	GPT-4o
ProposalReviewer	Quality-check proposal	proposal.md	quality_score + suggested edits	GPT-4o

Customer Success Swarm (3 agents)

Agent	Task	Input	Output	Model
ExpansionSignalDetector	Detect expansion opportunities	customer usage + signals	expansion_opportunity	GPT-4o-mini
QBRPreparer	Prepare quarterly business review	customer data + outcomes	qbr_deck.md	GPT-4o
RenewalRiskScorer	Score renewal risk	customer engagement	renewal_risk_score	GPT-4o-mini

Finance Swarm (3 agents)

Agent	Task	Input	Output	Model
TokenCostTracker	Track token spend per agent	agent_runs table	cost_report.json	GPT-4o-mini
ROICalculator	Calculate agent ROI scores	cost + revenue_influenced	roi_score per agent	GPT-4o-mini
BudgetAlerter	Alert on budget overruns	daily spend vs budget	alert if > threshold	GPT-4o-mini

Governance Swarm (3 agents)

Agent	Task	Input	Output	Model
OutreachAuditor	Audit all outreach before approval	outreach_campaigns	audit_flag + notes	GPT-4o
DataQualityMonitor	Monitor data quality across tables	all tables	quality_report.json	GPT-4o-mini
ComplianceChecker	Check GDPR/CAN-SPAM compliance	outreach content + contact	compliance_pass/fail	GPT-4o-mini

4.3 Agent ROI Formula

```
Agent ROI Score = Revenue Influenced (USD) / Cost Generated (USD)

Target ROI: > 100x (i.e., every $1 of agent cost should influence $100+ of revenu
Warning threshold: < 50x
Shutdown threshold: < 20x (after 30-day evaluation period)
```

Revenue influenced is attributed using last-touch attribution: the agent whose output was most recently used in a won opportunity receives full revenue attribution. This is a simplification — a more sophisticated multi-touch attribution model is recommended at 50+ customers.

PART 5 — WORKFLOW MAPS

5.1 Primary Revenue Workflow

```
TRIGGER: Daily scheduled run (06:00 UTC)
│
├── [S1: Discovery] SectorScanner runs for each target sector × region combinatic
│   ├── Output: 50-200 new company candidates per run
│   ├── CompanyProfiler enriches each candidate
│   ├── AISignalDetector scans for signals
│   ├── OpportunityScorer scores each company
│   └── QualityGatekeeper filters: only companies scoring ≥ 60 proceed
│
├── [Human Gate 1] Account Director reviews new companies ≥ 80 score
│   └── Approved companies enter Intelligence queue
│
├── [S2: Intelligence] Intelligence Swarm builds dossier
│   ├── All 6 agents run sequentially
│   ├── DossierAssembler produces account_dossier_[company].md
│   └── Dossier stored in S3 + database
│
├── [Human Gate 2] Account Director reviews dossier
│   └── Approved dossiers enter Decision Detection queue
│
├── [S3: Decision Detection] Decision Swarm monitors for triggers
│   ├── NewsMonitor runs every 6 hours per tracked company
│   ├── DecisionClassifier scores each news item
│   └── High-confidence decisions (≥ 70) trigger Outreach queue
│
├── [S4: Outreach Factory] Outreach Swarm generates campaign
│   ├── IntroEmailWriter personalises based on dossier + decision
│   ├── LinkedInMessageWriter generates connection note
│   ├── FollowUpSequenceWriter prepares follow-ups 1 and 2
│   └── All outputs enter approval queue with status = pending_approval
│
├── [Human Gate 3] Account Director reviews and approves outreach
│   └── Approved outreach is sent (manually or via CRM integration)
│
└── [Outcome Tracking] Replies, meetings, opportunities updated in CRM
    └── Outcomes feed Learning Feedback Loop
```

5.2 Proposal Workflow

```
TRIGGER: Meeting booked (opportunity.stage = meeting_booked)
│
├── [Proposal Swarm] UseCaseSelector identifies top 3 Mesh use cases
├── [Proposal Swarm] PricingModeller generates 3 pricing options
├── [Proposal Swarm] ProposalWriter generates full proposal
├── [Proposal Swarm] ProposalReviewer quality-checks
│
├── [Human Gate 4] Account Director reviews proposal
│   └── Approved proposals delivered to prospect
│
└── [Outcome Tracking] Proposal outcome updated in pipeline
```

5.3 Token Throttling Workflow

```
TRIGGER: Hourly agent ROI check (Finance Swarm)
│
├── ROICalculator computes rolling 7-day ROI for each agent
├── BudgetAlerter checks daily spend vs budget
│
├── If agent ROI < 50x for 7 days:
│   └── Agent flagged for review → AI Operations Lead notified
│
├── If agent ROI < 20x for 30 days:
│   └── Agent automatically throttled (50% capacity reduction)
│   └── agent_throttle_log entry created
│
└── If daily token spend > 120% of budget:
    └── All non-critical agents paused until next day
    └── CEO and AI Operations Lead notified
```

5.4 Learning Feedback Workflow

```
TRIGGER: Weekly (Sunday 23:00 UTC)
│
├── Analyse won opportunities: sector, region, decision_type, dossier_score
├── Analyse lost opportunities: same dimensions
├── Identify best-performing outreach sequences by response rate
├── Identify best-performing Decision Twin patterns by engagement probability
│
├── Update targeting criteria:
│   └── Increase weight for high-win sectors/regions
│   └── Decrease weight for low-win sectors/regions
│
└── Generate weekly performance report → Revenue Command Center
```

PART 6 — DASHBOARD SPECIFICATIONS

6.1 Revenue Command Center — Layout

The Revenue Command Center is a single-page executive dashboard accessible at `/revenue-command` (authenticated, CEO and Account Directors only). It has four panels:

Panel 1 — Pipeline Funnel (top left)

A live funnel showing counts at each stage:

Stage	Count	Week-over-Week Change
Companies Discovered	0	—
Dossiers Completed	0	—
Decisions Detected	0	—
Outreach Queued	0	—
Outreach Sent	0	—
Meetings Booked	0	—
Opportunities Created	0	—
Proposals Delivered	0	—
Customers Won	0	—

Panel 2 — Financial Metrics (top right)

Metric	Current	Target	Status
ARR	$0	$20M	🔴
ACV Forecast	$0	$667K avg	🔴
Pipeline Value	$0	$60M	🔴
Win Rate	—	15%	—
CAC Estimate	—	<$50K	—
Revenue per Employee	—	$2.5M	—

Panel 3 — Operations (bottom left)

Metric	Current	Target	Status
Agent Utilisation	—	>80%	—
Daily Token Spend	$0	<$500	🟢
Monthly Azure Spend	$0	<$10K	🟢
Cost per Opportunity	—	<$200	—
Cost per Customer	—	<$50K	—
Agents Below ROI	0	0	🟢

Panel 4 — Approval Queues (bottom right)

Live count of items awaiting human review:

- Outreach pending approval: 0
- Dossiers pending review: 0
- Proposals pending review: 0
- Opportunities pending qualification: 0

6.2 Account Director View

Account Directors see a filtered view showing only their assigned accounts, with:

- Company list with opportunity score, stage, and last activity
- Outreach approval queue (their accounts only)

- Meeting calendar integration
- Proposal status tracker

6.3 AI Operations View

The AI Operations Lead sees:

- Agent run log (last 24 hours, last 7 days)
- Agent ROI scores (all agents, ranked)
- Throttled agents list
- Token spend by swarm
- Error rate by agent
- Queue depths (pending tasks per agent)

[End of Parts 1–6]

PART 7 — COST MODEL

7.1 Cost Structure

The AROS has three cost categories: infrastructure (Azure), LLM tokens, and human operators. At scale, the dominant cost is human operators — not technology.

Cost Category	Month 1	Month 6	Month 12	Month 24
Azure Infrastructure	$1,800	$4,500	$9,900	$15,000
LLM Tokens (OpenAI)	$1,200	$6,000	$12,000	$18,000
Human Operators (8 FTE)	$120,000	$120,000	$140,000	$160,000
Legal & Compliance	$5,000	$3,000	$3,000	$3,000
Sales & Marketing	$10,000	$20,000	$30,000	$40,000
Total Monthly OpEx	**$138,000**	**$153,500**	**$194,900**	**$236,000**
Annual OpEx	**$1.66M**	—	**$2.34M**	**$2.83M**

7.2 Unit Economics

Metric	Target	Notes
Cost per company discovered	$0.50	Primarily token cost
Cost per dossier completed	$8.00	6 GPT-4o calls per dossier
Cost per decision detected	$1.50	News monitoring + classification
Cost per outreach campaign	$3.00	5 email/LinkedIn variants
Cost per proposal	$15.00	Full proposal generation
Cost per opportunity created	150–250	All upstream costs amortised
Cost per customer acquired (CAC)	30,000–50,000	Including human operator time

7.3 Gross Margin Model

At 20MARRwith2.83M annual OpEx:

Metric	Value
ARR	$20,000,000
Annual OpEx	$2,830,000
Gross Profit	$17,170,000
Gross Margin	**85.9%**

This meets the 85%+ gross margin target. The model is sensitive to human operator headcount — adding a 9th operator at $160K/year reduces gross margin by approximately 0.8%.

7.4 Cost Scaling Triggers

Three events trigger a cost review:

1. **Monthly token spend exceeds $15,000** — review agent ROI scores; throttle underperformers
2. **Monthly Azure spend exceeds $12,000** — review database query patterns; consider read replicas
3. **CAC exceeds $75,000** — review outreach conversion rates; adjust targeting criteria

PART 8 — TOKEN MODEL

8.1 Token Budget Architecture

The token budget is allocated by swarm, not by individual agent. This prevents any single agent from consuming disproportionate resources.

Swarm	Monthly Token Budget	Estimated Cost	Priority
Intelligence Swarm	8,000,000 tokens	$8,000	P1 — revenue-critical
Outreach Swarm	4,000,000 tokens	$4,000	P1 — revenue-critical
Proposal Swarm	2,000,000 tokens	$2,000	P1 — revenue-critical
Discovery Swarm	3,000,000 tokens	$900	P2 — uses mini model
Decision Swarm	2,000,000 tokens	$600	P2 — uses mini model
Customer Success Swarm	1,000,000 tokens	$300	P2
Finance Swarm	500,000 tokens	$75	P3 — uses mini model
Governance Swarm	1,000,000 tokens	$300	P3
Total	**21,500,000 tokens**	**$16,175**	—

8.2 Model Routing Rules

The model routing decision is made at the agent level, not the swarm level. The routing logic is:

- **GPT-4o-mini** ($0.15/1Minput$,0.60/1M output): classification, scoring, deduplication, simple extraction, monitoring
- **GPT-4o** ($5.00/1Minput$,15.00/1M output): dossier writing, Decision Twin construction, proposal generation, outreach personalisation

The 15x cost difference between models means that routing a classification task to GPT-4o instead of GPT-4o-mini is a 15x cost error. The Governance Swarm audits model routing weekly.

8.3 Token Efficiency Targets

Agent Type	Target Tokens per Task	Maximum Tokens per Task
Company profiling	1,500	3,000
Signal detection	800	1,500
Dossier section (per section)	2,500	4,000
Full dossier assembly	500	1,000
Intro email	1,200	2,000
LinkedIn message	300	500
Full proposal	8,000	12,000
Decision brief	1,500	2,500

8.4 Token ROI Tracking

Every agent run records:

```
token_roi = revenue_influenced_usd / cost_usd
```

A dossier that costs $8togenerateandcontributestoa$200,000 ACV win has a token ROI of 25,000x. A dossier that costs $8 and contributes to nothing has a token ROI of 0.

The Finance Swarm calculates rolling 7-day and 30-day token ROI for every agent. Agents below 20x ROI for 30 consecutive days are automatically throttled.

PART 9 — REVENUE MODEL

9.1 Pricing Architecture

AgenThink Mesh offers three commercial tiers:

Tier	Name	ACV	Target Buyer	Scope
Tier 1	Retrospective Pilot	$25,000$–75,000	Innovation / R&D leads	Single case, 30 days, proof of concept
Tier 2	Enterprise Annual	$150,000$–500,000	CDO / CTO / Head of Strategy	Unlimited cases, 12 months, full platform
Tier 3	Enterprise Platform	$500,000$–2,000,000	C-Suite	Custom deployment, dedicated agents, SLA

9.2 Revenue Scenarios

Conservative Scenario (Year 1–3)

Year	Pilots	Enterprise Annual	Enterprise Platform	ARR
1	5 × $50K	2 × $200K	0	$650K
2	8 × $50K	6 × $250K	1 × $750K	$2,650K
3	10 × $50K	12 × $300K	3 × $1M	$7,100K

Base Scenario (Year 1–5)

Year	Pilots	Enterprise Annual	Enterprise Platform	ARR
1	8 × $50K	3 × $200K	0	$1,000K
2	12 × $50K	8 × $250K	2 × $750K	$3,100K
3	15 × $50K	15 × $300K	5 × $1M	$10,250K
4	10 × $50K	20 × $350K	8 × $1.2M	$17,100K
5	8 × $50K	22 × $400K	10 × $1.5M	$24,200K

Aggressive Scenario (Year 1–4)

Year	Pilots	Enterprise Annual	Enterprise Platform	ARR
1	12 × $50K	5 × $250K	0	$1,850K
2	15 × $50K	12 × $300K	3 × $1M	$7,350K
3	10 × $50K	18 × $350K	7 × $1.2M	$15,200K
4	8 × $50K	20 × $400K	12 × $1.5M	$26,400K

9.3 Revenue per Human Operator

At $20MARRwith8operators : **$2.5M revenue per human operator**. This is the primary metric that justifies the autonomous architecture investment. A traditional enterprise software company with 8 salespeople generates approximately $800K–$1.2M per rep. The AROS target is 2–3x higher because agents handle all discovery, intelligence, and outreach preparation.

9.4 Pipeline Velocity Model

To reach $20M ARR, the pipeline must maintain:

Stage	Required Count	Conversion Rate	Upstream Required
Customers Won	30	—	—
Proposals Delivered	200	15%	—
Meetings Booked	600	33%	—
Outreach Sent	6,000	10%	—
Dossiers Completed	12,000	50%	—
Companies Discovered	100,000	12%	—

At 100 companies discovered per day, the 100,000 company index is reached in approximately 3 years of continuous operation.

PART 10 — GOVERNANCE MODEL

10.1 Governance Principles

The AROS operates under four governance principles:

1. **No agent sends outreach without human approval.** The approval gate is non-negotiable and cannot be bypassed by any agent or automated workflow.

2. **All agent outputs are auditable.** Every agent run is logged with full input, output, token count, cost, and timestamp. Logs are retained for 7 years.
3. **All outreach is GDPR and CAN-SPAM compliant.** The Governance Swarm checks every outreach item before it enters the approval queue.
4. **All data is classified.** Company data, contact data, and dossiers are classified as Confidential. Customer data is classified as Restricted.

10.2 Human Approval Gates

Gate	Trigger	Approver	SLA	Override
Gate 1: Company qualification	Opportunity score ≥ 80	Account Director	24 hours	CEO only
Gate 2: Dossier approval	Dossier completed	Account Director	48 hours	CEO only
Gate 3: Outreach approval	Outreach generated	Account Director	24 hours	None — no override
Gate 4: Proposal approval	Proposal generated	Account Director + CEO	48 hours	None — no override

Gate 3 has no override. No outreach is ever sent without explicit human approval. This is the primary compliance and reputational protection mechanism.

10.3 Data Governance

Data Type	Classification	Retention	Access
Company profiles	Internal	5 years	All operators
Contact data	Confidential	3 years (GDPR)	Account Directors + CEO
Dossiers	Confidential	5 years	Account Directors + CEO
Outreach content	Confidential	7 years	Account Directors + CEO
Customer data	Restricted	7 years	CEO + Finance Lead
Agent run logs	Internal	7 years	AI Operations + CEO
Token cost data	Internal	3 years	Finance Lead + CEO

10.4 Compliance Framework

The AROS is designed for compliance with:

- **GDPR** (EU/UK): Right to erasure implemented via `DELETE FROM contacts WHERE id = ?` with cascade. Data processing agreements required before storing any EU/UK contact data.
- **CAN-SPAM** (US): All outreach includes unsubscribe mechanism. Unsubscribes are processed within 10 business days.
- **CASL** (Canada): Explicit consent required before contacting Canadian contacts. Consent records stored in `contacts.email_verified` field.
- **PDPA** (Singapore): Data localisation requirements reviewed before storing Singapore contact data.

10.5 Agent Governance Rules

Rule	Description	Enforcement
AG-001	No agent may send external communications	Technical block at API gateway
AG-002	No agent may access customer data without explicit authorisation	Database row-level security
AG-003	All agent outputs must be written to database before downstream use	Enforced by orchestration framework
AG-004	Agents below 20x ROI for 30 days are automatically throttled	Finance Swarm automated enforcement
AG-005	All outreach must pass Governance Swarm compliance check	Enforced before approval queue entry
AG-006	No agent may modify its own configuration	Technical block
AG-007	All agent runs must record token count and cost	Enforced at SDK level

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PART 11 — OPERATING MANUAL

11.1 Daily Operations (8 Human Operators)

CEO (1 hour/day)

- Review Revenue Command Center dashboard

- Review and sign off on proposals > $500K
- Review weekly learning report
- Approve any Gate 4 proposals

CTO (2 hours/day)

- Review agent error logs
- Review token spend vs budget
- Approve any infrastructure scaling decisions
- Review Governance Swarm audit flags

Product Lead (2 hours/day)

- Review dossier quality samples (5 per day)
- Review outreach quality samples (10 per day)
- Flag quality issues to AI Operations Lead
- Manage product roadmap

Account Director Americas (4 hours/day)

- Process Gate 1 approvals (company qualification)
- Process Gate 2 approvals (dossier review)
- Process Gate 3 approvals (outreach review)
- Manage active opportunities
- Conduct meetings with prospects

Account Director EMEA/APAC (4 hours/day)

- Same as Account Director Americas for EMEA/APAC accounts

AI Operations Lead (6 hours/day)

- Monitor agent performance
- Investigate agent failures
- Tune agent prompts and parameters
- Manage agent throttling
- Coordinate with CTO on infrastructure

Platform Engineer (6 hours/day)

- Maintain Azure infrastructure

- Deploy agent updates
- Monitor system health
- Manage database performance

Finance & Compliance Lead (3 hours/day)

- Review daily token spend report
- Review agent ROI scores
- Process GDPR deletion requests
- Manage legal and compliance documentation

11.2 Weekly Cadence

Day	Activity	Owner
Monday	Pipeline review meeting (30 min)	CEO + Account Directors
Tuesday	Agent performance review (30 min)	CTO + AI Operations Lead
Wednesday	Dossier quality review (1 hour)	Product Lead + Account Directors
Thursday	Outreach conversion review (30 min)	Account Directors + AI Operations
Friday	Weekly learning report review (30 min)	CEO + all
Sunday	Learning Feedback Workflow runs (automated)	System

11.3 Escalation Paths

Event	First Response	Escalation	Time Limit
Agent failure	AI Operations Lead	CTO	2 hours
Data breach	Finance & Compliance Lead	CEO + Legal	1 hour
GDPR deletion request	Finance & Compliance Lead	CEO	72 hours
Budget overrun (>120%)	Finance & Compliance Lead	CEO	Same day
Outreach compliance flag	Account Director	Finance & Compliance Lead	24 hours
Customer complaint	Account Director	CEO	4 hours

11.4 Onboarding New Operators

When a new operator joins (within the 8-person limit):

1. Complete data governance training (2 hours)
2. Complete GDPR/CAN-SPAM compliance training (1 hour)
3. Complete Revenue Command Center orientation (1 hour)
4. Shadow existing Account Director for 5 days
5. Process first 10 approvals under supervision
6. Independent operation begins at day 10

PART 12 — EXPANSION ROADMAP

12.1 Phase 1 — Foundation (Months 1–6)

Goal: First 5 paying customers. Validate the core workflow.

Milestone	Target	Month
AROS infrastructure deployed	Azure environment live	Month 1
Discovery Swarm operational	100 companies/day	Month 1
Intelligence Swarm operational	10 dossiers/day	Month 2
First dossier approved	1 dossier	Month 2
First outreach sent	1 campaign	Month 3
First meeting booked	1 meeting	Month 3–4
First pilot signed	$25K$–75K	Month 4–5
5 pilots signed	$125K$–375K ARR	Month 6

12.2 Phase 2 — Scale (Months 7–18)

Goal: First 15 enterprise customers. $3M ARR.

Milestone	Target	Month
Decision Detection Engine live	50 decisions/week	Month 7
Outreach Factory fully automated	100 campaigns/week	Month 8
First enterprise annual contract	$150K$–500K	Month 9
10 enterprise customers	$2M ARR	Month 12
Customer Success Swarm live	Expansion pipeline	Month 13
First renewal	90%+ retention	Month 15
15 enterprise customers	$3M ARR	Month 18

12.3 Phase 3 — Dominance (Months 19–36)

Goal: 30+ enterprise customers. $20M ARR.

Milestone	Target	Month
100,000 companies indexed	Full target universe	Month 24
10,000 executive profiles	Full contact database	Month 24
1,000 active opportunities	Full pipeline	Month 24
20 enterprise customers	$8M ARR	Month 24
Proposal Swarm fully automated	20 proposals/week	Month 25
Learning model v2	Improved targeting	Month 28
30 enterprise customers	$15M ARR	Month 30
$20M ARR	**Target achieved**	**Month 36**

12.4 Geographic Expansion Sequence

The AROS expands geographically in a specific sequence designed to maximise early wins:

1. **United States** (Month 1) — largest market, most AI-native buyers, English-language content
2. **United Kingdom** (Month 3) — English-language, strong financial services sector
3. **Singapore** (Month 6) — APAC gateway, high AI adoption, English-language
4. **UAE** (Month 9) — existing relationships from Diriyah/GCC work, sovereign investors
5. **Australia** (Month 12) — English-language, growing AI adoption
6. **Canada** (Month 15) — English-language, CASL compliance required
7. **Hong Kong** (Month 18) — financial services hub
8. **Saudi Arabia** (Month 21) — Vision 2030 digital transformation pipeline
9. **Japan** (Month 24) — largest APAC market; requires Japanese-language capability

12.5 Sector Expansion Sequence

Priority	Sector	Rationale	Target Month
1	Banking	Highest AI adoption, highest ACV	Month 1
2	Asset Management	Decision Twin use case strongest	Month 1
3	Insurance	High data complexity, strong governance need	Month 3
4	Telecom	Large digital transformation budgets	Month 6
5	Energy & Utilities	Infrastructure investment wave	Month 9
6	Logistics	Supply chain AI adoption	Month 12
7	Infrastructure	Government-adjacent, long sales cycles	Month 15
8	Sovereign Investors	Highest ACV potential; longest sales cycle	Month 18
9	Government	Regulatory complexity; separate compliance track	Month 24

12.6 Agent Count Expansion

Phase	Agent Count	Swarms Active	Monthly Token Budget
Foundation (M1–6)	34 agents	Discovery + Intelligence + Outreach	$5,000
Scale (M7–18)	120 agents	All 8 swarms	$12,000
Dominance (M19–36)	200+ agents	All 8 swarms + specialised variants	$18,000

FINAL SUCCESS CRITERIA

The AROS is complete when AgenThink Mesh can perform all of the following without human initiation:

Capability	System	Status
Discover opportunities autonomously	S1: Discovery Engine	Architecture complete
Build account intelligence autonomously	S2: Intelligence Factory	Architecture complete
Detect strategic decisions autonomously	S3: Decision Detection	Architecture complete
Generate outreach autonomously (pending approval)	S4: Outreach Factory	Architecture complete
Generate proposals autonomously (pending approval)	S4: Proposal Swarm	Architecture complete
Track pipeline autonomously	S5: Revenue Command Center	Architecture complete
Optimise token spend autonomously	Finance Swarm	Architecture complete
Support 30+ enterprise customers with 8 humans	All systems	Target: Month 36

APPENDIX — IMPLEMENTATION PRIORITY ORDER

The following sequence is recommended for implementation. Each item is a discrete deliverable that can be built and tested independently.

Priority	Deliverable	Estimated Effort	Dependencies
P0-1	Database schema deployment	2 days	Azure PostgreSQL
P0-2	Azure infrastructure provisioning	3 days	Azure subscription
P0-3	Agent orchestration framework	5 days	Database + Azure Container Apps
P0-4	Discovery Swarm (8 agents)	5 days	Orchestration framework
P0-5	Revenue Command Center (basic)	3 days	Database
P1-1	Intelligence Swarm (6 agents)	5 days	Discovery Swarm
P1-2	Outreach Factory (5 agents)	5 days	Intelligence Swarm
P1-3	Human approval queue UI	3 days	Outreach Factory
P1-4	Finance Swarm (3 agents)	2 days	Agent runs table
P1-5	Governance Swarm (3 agents)	3 days	Outreach Factory
P2-1	Decision Detection Engine (5 agents)	5 days	Intelligence Swarm
P2-2	Proposal Swarm (4 agents)	5 days	Intelligence Swarm
P2-3	Customer Success Swarm (3 agents)	3 days	Pipeline table
P2-4	Learning Feedback Workflow	4 days	All swarms
P2-5	Revenue Command Center (full)	5 days	All systems

Total estimated implementation effort: 63 engineering days (approximately 3 months with 1 Platform Engineer).

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